

Numerical Verification Supplement

1 Numerical Diagnostics and Galerkin Verification

This supplement records finite-dimensional numerical diagnostics for the calibrated shorting mechanism. It includes only the verification material tied to the paper’s Galerkin consistency result, the exact calibration result, the finite-part regime separation, and the admissibility boundary. The proof appendix does not depend on these computations.

Verification Map

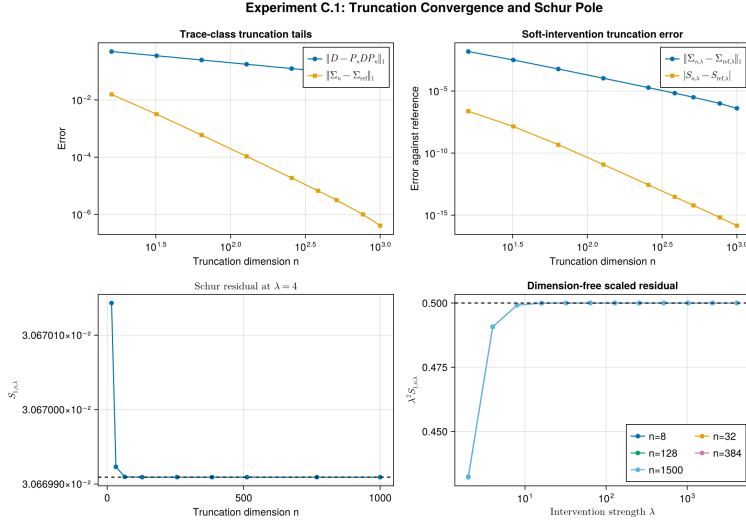
- Section 1.1: Galerkin truncation and variational shorting consistency.
- Section 1.2: canonical variance scaling.
- Section 1.3: finite-part regime separation in a three-node Gaussian chain.
- Section 1.4: admissibility boundary stress test.
- Section 1.5: sparse OU/GFF precision-operator implementation and comparison.
- Section 1.6: BigFloat/Float64 agreement audit for residual magnitudes.

1.1 Galerkin Verification and Ill-Conditioning

We illustrate the Galerkin convergence asserted by the paper’s Galerkin consistency result against a finite reference on an admissible model. The Hilbert space is $\ell^2(\mathbb{N})$ with coordinate basis $\{\phi_j\}$. The drift is $\mathcal{A} = \mathcal{A}_0 + \mathcal{K}$, where $\mathcal{A}_0\phi_j = -(1 + 0.01j^2)\phi_j$ and $\mathcal{K}\phi_1 = \sum_{j \geq 2} 12j^{-3}\phi_j$ is a bounded rank-one nilpotent perturbation. The noise is $\mathcal{D}\phi_j = j^{-3/2}\phi_j$, which is trace-class. The regularized conditioning acts at $k = 1$ with $\varepsilon_0 = 1$. Truncated covariances are compared against a finite reference $N_{\text{ref}} = 1500$.

Table 1: Truncation convergence diagnostics.

Diagnostic	Value	Expected
$\ \mathcal{D} - \Pi_n \mathcal{D} \Pi_n\ _{\mathcal{S}_1}$ slope	-0.499	$\rightarrow 0$
$\ \Sigma_n - \Sigma_{\text{ref}}\ _{\mathcal{S}_1}$ slope	-2.683	$\rightarrow 0$
$\ \Sigma_{n,\lambda} - \Sigma_{\text{ref},\lambda}\ _{\mathcal{S}_1}$ slope	-2.683	$\rightarrow 0$
$ S_{1,n,\lambda} - S_{1,\text{ref},\lambda} $ slope	-5.504	$\rightarrow 0$
$m_2^{\text{Schur},(1000)}$ extrapolated	0.5000000	0.5
$ m_2^{\text{Schur},(1000)} - \varepsilon_0/2 $	6.87×10^{-14}	0


 Figure 1: Trace-norm and residual errors decay in n ; the extrapolated $m_2^{\text{Schur},(1000)}$ differs from $\varepsilon_0/2$ by 6.87×10^{-14} .

The variational operator sequence converges, but computing the truncated Schur complement by naive inversion,

$$S^{(n)} = \Sigma_{EE} - \Sigma_{EI}^{(n)} (\Sigma_{II}^{(n)})^{-1} \Sigma_{IE}^{(n)},$$

fails in generic Galerkin bases. The internal block is a trace-class covariance, so its truncated matrix representation becomes severely ill-conditioned as n grows. The diagnostics show that for $n \geq 400$ the condition number $\kappa(\Sigma_{II}^{(n)})$ grows as n^6 and formally exceeds the 10^{16} IEEE 754 Float64 precision limit.

1.2 Canonical Scaling Diagnostic

We validate the canonical normalization in the paper’s exact calibration result through its decoupled reference channel. With incoming target-coordinate blocks and off-diagonal drift coupling set to zero, the target covariance is exactly

$$\Sigma_{\lambda,22} = d_{\lambda}/(2\lambda).$$

Four candidate target-diffusion families are tested:

1. **Canonical:** $d_{\lambda} = \varepsilon_0/\lambda$.
2. **Slow:** $d_{\lambda} = \varepsilon_0$.
3. **Fast:** $d_{\lambda} = \varepsilon_0/\lambda^2$.
4. **Perturbed:** $d_{\lambda} = (\varepsilon_0/\lambda)(1 + \lambda^{-2})$.

Table 2: Exact-calibration ablation. Only the canonical family satisfies the exact finite- λ covariance condition.

Family	$\lambda d_{\lambda}/\varepsilon_0$	defect $ 2\lambda^2 \Sigma_{\lambda,kk}/\varepsilon_0 - 1 $
Canonical	1	0
Slow	λ	$\lambda - 1 \rightarrow \infty$
Fast	λ^{-1}	$ 1 - \lambda^{-1} \rightarrow 1$
Perturbed	$1 + \lambda^{-2}$	λ^{-2}

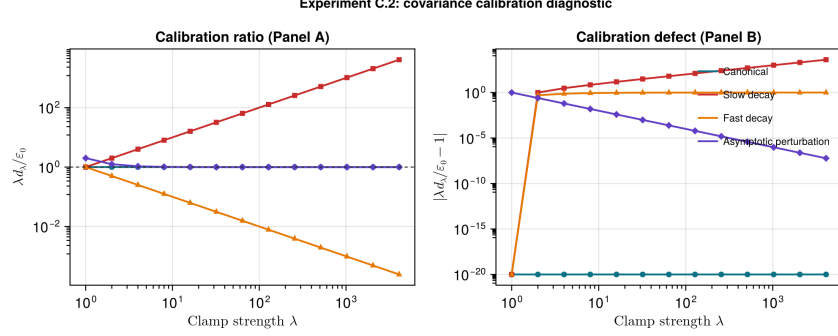


Figure 2: The perturbed family has the correct limiting ratio but fails the exact finite- λ covariance condition. Exact zero defects are displayed at 10^{-20} .

1.3 Three-Node Gaussian Chain

We evaluate the Gaussian Fredholm finite-part functional $\mathfrak{J}_\lambda$ on a three-node linear Ornstein–Uhlenbeck SDE,

$$X_1 \rightarrow X_2 \rightarrow X_3,$$

where X_2 is the target coordinate subject to regularized conditioning of strength $\lambda \in \{2^1, \dots, 2^{12}\}$, with $x_2^* = 0$ and $\varepsilon_0 = 1$. The drift and diffusion are

$$A_\lambda = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & a_{32} & -1 \end{pmatrix}, \quad D_\lambda = \text{diag}(1, \varepsilon_0/\lambda, 1).$$

The target mean forcing b_2 controls the target mean shift $m_{\lambda,2} = b_2/\lambda$.

1. **Case A, decoupled-centered:** $a_{32} = 0$, $b_2 = 0$, and $\mathfrak{J}_\lambda = 0$ exactly.
2. **Case B, coupled-centered:** $a_{32} = 1$, $b_2 = 0$, and $\mathfrak{J}_\lambda = O(\lambda^{-4}) \rightarrow 0$.
3. **Case C, coupled-off-centered:** $a_{32} = 1$, $b_2 = 1$, and $\mathfrak{J}_\lambda \rightarrow b_2^2/\varepsilon_0 = 1$.

1.4 Cameron–Martin Admissibility Stress Test

For a 1D elliptic Gaussian field, smooth compactly supported sensors have finite Cameron–Martin energy. Approximating a point sensor by a concentrated mollifier g_ε drives the energy norm to infinity as $\varepsilon \downarrow 0$. This stress

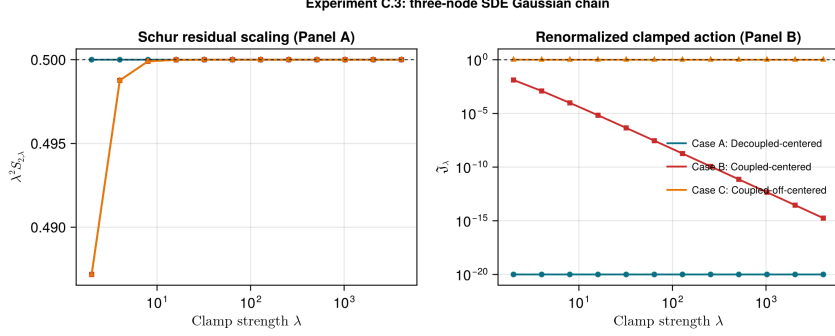


Figure 3: The panels show $\lambda^2 S_{2,\lambda} \rightarrow 0.5$ and the three finite-part regimes: exact zero, $O(\lambda^{-4})$ decay, and convergence to 1.

test illustrates the admissibility boundary in the paper’s theorem: smoothed observations are admissible, while the unsmoothed Dirac point-evaluation limit is not a finite-energy channel for the shorting theorem.

1.5 Sparse Precision-Operator Implementation

The diagnostics also include the OU framework with drift $\mathcal{A}_0 = -L^{-1}$ and formal covariance $Q_0 = L$ on the dual space $H^{-1}(0, 1)$. The resolvent admits the sparse conjugate realization

$$(\lambda I + L^{-1})^{-1} = L(\lambda L + I)^{-1} = \frac{1}{\lambda} L(L + \tilde{\lambda} I)^{-1}, \quad \tilde{\lambda} = \lambda^{-1}.$$

Thus the resolvent is obtained by a single sparse tridiagonal solve with the precision operator L at the conjugate parameter $\tilde{\lambda}$, avoiding dense covariance construction.

Table 3: Sparse precision-operator validation in the OU regime, $N = 10000$, $\kappa = 1$, $\Delta_E = 1$.

λ	$\tilde{\lambda} = \lambda^{-1}$	$\lambda^4 \text{tr}(\rho_{\lambda,E})$
10^1	10^{-1}	1.0263×10^{-7}
10^2	10^{-2}	1.0283×10^{-7}
10^3	10^{-3}	1.0285×10^{-7}
10^4	10^{-4}	1.0285×10^{-7}
10^5	10^{-5}	1.0285×10^{-7}

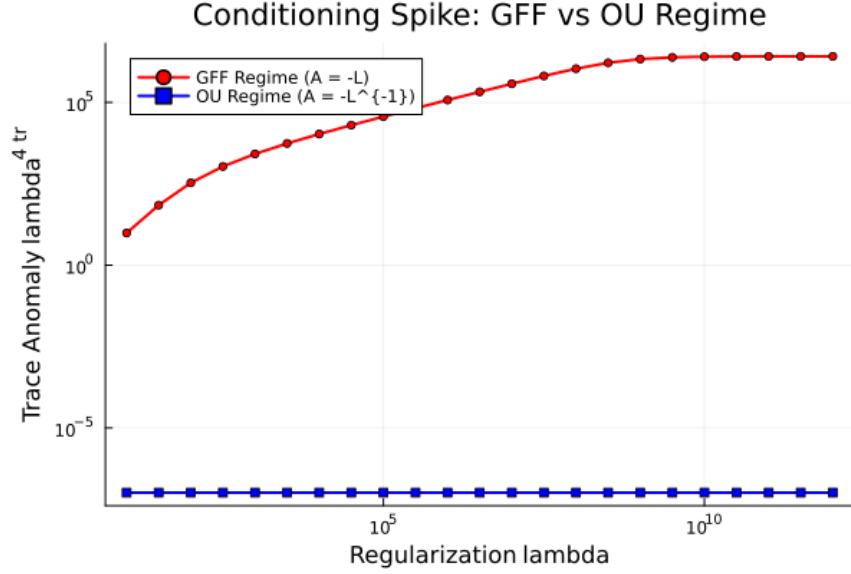


Figure 4: Trace anomaly conditioning comparison between the GFF and OU regimes.

1.6 Precision Audit

At $n = 256$, representative Schur residuals were recomputed using IEEE 754 Float64 and 256-bit Julia BigFloat arithmetic with the same covariance equation and Schur assembly.

Table 4: Precision audit at $n = 256$; S_{256} uses 256-bit BigFloat.

λ	$\kappa_2(\Sigma_{II})$	$ \lambda^2 S - \frac{1}{2} $	$ S_{64} - S_{256} / S_{256} $
4	2.97×10^6	3.45×10^{-1}	6.50×10^{-16}
64	9.14×10^5	1.13×10^{-3}	3.48×10^{-17}
4096	9.14×10^5	9.20×10^{-10}	5.54×10^{-17}

For the five-point extrapolation of $m_2^{\text{Schur},(256)}$, Float64 and BigFloat differ by 9.98×10^{-17} ; the finite- λ remainder, not rounding, governs the observed distance to the limiting constant.